Module 09 — Lab: CRUD + Auth Consolidation

Phase: Backend (protected lab — freeze the API)
Previous module: Spring Security Lite
Next module: Lab: Deploy + Capstone Debug
Lab kit: 15_Lab_Assets/19_spring_auth_tasks/

Module Outcome:

- Freeze the API: 401 before 400, owner rules, statuses a demo can show
- Break a classmate's auth on purpose — then fix yours
- Leave a demo bundle (Postman + README) a recruiter could follow

Teach order today: what consolidation *is* → end-to-end glue path → feature freeze IN/OUT → must-pass statuses (401 before 400) → peer break-auth → demo bundle.

Prerequisites

- **CampusCart JWT lite** from Spring Security Lite plus catalog DTOs / `@Valid` / 400 / 404 from modules 03 and 04. Kit 19. JDK 17 + Maven 3.9+ + Spring Boot 3.x / `jakarta.*` (Tested Toolchain).
- **Postman** kit 19 collection. This lab **extends** the same file for freeze 400 / 404 — it does not start a new Boot app in git.

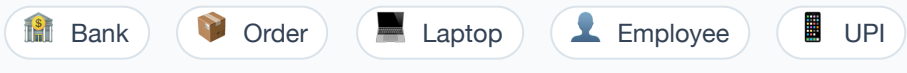
Definition / Purpose

Definition: Consolidation means one **running** API where already-taught pieces **fit**. CampusCart is not three homework folders. It is **one** MySQL catalog with DTO JSON, lite errors, and a JWT door on **writes**.

Definition: A **feature freeze** is a locked **IN** list for demo day. Anything not on the list is **OUT** — even if it would look impressive. Freeze is how a batch **finishes**.

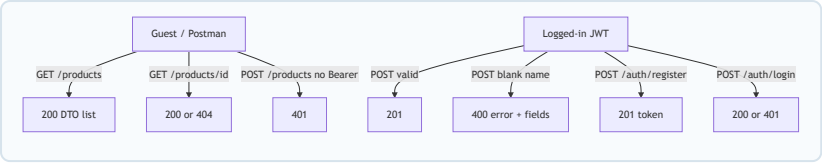
Definition: Peer break-auth is a review drill: a partner (or kit `broken/`) ships an API that **looks** logged-in. You **name the smell** and the **fix** already taught in module 05. You do not invent OAuth2, XSS chapters, or new filters.

Purpose: Priya's JWT homework and Ravi's DTO catalog lived in different clones. Demo day needs **one** Postman run: guest **GET**s Pani puri; a logged-in user **POST**s it; blank name is **400**; missing id is **404**; no token is **401**. Interviews will ask "walk me through the API," not "recite `BCryptPasswordEncoder`."



Where used?
- CampusCart — college Swiggy-style menu: public read, protected create
- Swiggy — catalog freeze before a storefront (React is **Later —**)
- HDFC Bank — prove identity before a write (**no money movement here**)
- Employee Management Portal — staff login before a leave POST (**Later —** roles)
- Apollo Hospitals — public doctor list; booking needs a token (**Later —**)

Visual — freeze path (one API):



Key takeaways:

- Consolidation **glues** modules 03–05 — it does not reteach them
- **Freeze IN** is the demo contract; **OUT** is not failure, it is later
- Security runs **before** `@Valid` — no token → **401**, not 400
- Kit 19 `broken/` is the break-auth lab — four smells, four fixes
- Secrets stay in **env**; the bundle is README + collection + screenshots

Motivation — Feel the pain of two half-APIs

Ravi's clone returns 201 on POST with **no** header. Priya's clone hashes passwords but GET by id still dumps the entity. QA cannot replay one collection. Freeze day exists so the **same** laptop shows **both**.

```
public class TwoHalfApisPain {
    public static void main(String[] args) {
        System.out.println("Catalog without a door -> junk rows");
        System.out.println("JWT without DTOs -> table leaked as JSON");
        System.out.println("Freeze = one API that does both");
    }
}
```

Expected Output:

```
Catalog without a door -> junk rows
JWT without DTOs -> table leaked as JSON
Freeze = one API that does both
```

Remember: Kit 19 already has the pieces. This lab **proves** they still work together after you touch auth.

Interview: "Did you finish Spring Security?" is weaker than "Guest GET, Bearer POST, 400 with token, 401 without."

Important Concepts

1. What consolidation is (and is not)

Definition: Consolidation is **integration**. You already wrote `Product` + `ProductRequest` + `AppUser` + `SecurityFilterChain`. Today you **run the path** and **stop adding features**.

Purpose: Without a freeze, every learner starts DELETE, CORS, and "admin role" the night before demo. None of those are required to pass. The batch looks unfinished.

Consolidation is	Consolidation is not
One Boot app (kit 19)	A new kit 20
Point at notes 03–05	Rewrite BCrypt / JWT / JPA
Status proof in Postman	A second error-body textbook
Diagnose kit `broken/`	New attack / OAuth2 chapters
README + demo bundle	Deploy / Actuator (**Later —**)

```
public class ConsolidationIs {
    public static void main(String[] args) {
        System.out.println("Glue 03 + 04 + 05 in kit 19");
        System.out.println("Do not rewrite SecurityFilterChain");
    }
}
```

Expected Output:

```
Glue 03 + 04 + 05 in kit 19
Do not rewrite SecurityFilterChain
```

Note:
- Wrong: opening a blank `start.spring.io` project “to keep it clean.”
- Wrong: copying module 05’s JWT lesson into this file as homework.
- Prefer: clone / copy kit 19 `starter/` or fix `broken/` until freeze statuses match.

Common Error: Treating freeze as “add more endpoints.” Freeze is **lock** the IN list.

Remember: If a feature is not on the IN table, it is Stretch or **Later —**.

Interview: What does “protected lab” mean here? Time to **finish and demo**, not a new topic.

Print that this lab glues modules already taught: 03 JPA persist, 04 DTO Valid 400, 05 JWT BCrypt chain — it is not a rewrite.

```
public class GlueNotRewrite {
    public static void main(String[] args) {
        System.out.println("03 JPA persist – already taught");
        System.out.println("04 DTO Valid 400 – already taught");
        System.out.println("05 JWT BCrypt chain – already taught");
    }
}
```

Expected Output:

```
03 JPA persist – already taught
04 DTO Valid 400 – already taught
05 JWT BCrypt chain – already taught
```

Print that you use kit 19 `starter` or `broken`, and there is no kit 20.

```
public class OneKitPrint {
    public static void main(String[] args) {
        System.out.println("kit 19 starter or broken");
        System.out.println("no kit 20");
    }
}
```

Expected Output:

```
kit 19 starter or broken
no kit 20
```

---

2. End-to-end CampusCart path

Definition: The **happy path** is one story: MySQL holds snacks and users. JSON in/out is DTOs. `@Valid` rejects blank names. JWT register / login issues a Bearer token. **GET** is a public menu. **POST** is a write.

Purpose: A hiring manager will say “show me.” You open Postman and walk **left to right** — not “I have a SecurityConfig somewhere.”

Do not rewrite mapping, `JpaRepository`, `PasswordEncoder`, or `JwtService`. Open the files you already filled:

Piece	Already taught	Kit 19 type
`Product` + repo	Module 03	Filled
`ProductRequest` / `ProductResponse` + mapper	Module 04	Filled
`@Valid` + `@RestControllerAdvice` 400	Module 04	Filled
GET by id **404**	Modules 02–04	Filled
`AppUser`, register / login, BCrypt, JWT, chain	Module 05	TODOs on `starter/`; bugs on `broken/`

Drop-in POST (already in kit 19 — **do not replace** with entity JSON):

```
@PostMapping
@ResponseStatus(HttpStatus.CREATED)
public ProductResponse add(@Valid @RequestBody ProductRequest incoming) {
    Product saved = products.save(ProductMapper.toEntity(incoming));
    return ProductMapper.toResponse(saved);
}
```

Chain recap (module 05 — **do not rewrite**; confirm GET stays public):

```
.requestMatchers("/auth/**").permitAll()
.requestMatchers(HttpMethod.GET, "/products", "/products/**").permitAll()
.anyRequest().authenticated()
```

Note:
- Wrong: hiding GET `/products/{id}` behind JWT “to be extra safe” on the core path — the menu is public.
- Wrong: POSTing the `Product` entity so `id` / Hibernate fields leak.
- Prefer: DTO in, DTO out; GET public; POST authenticated.

Common Error: “Auth works” while GET by id still returns 500. Freeze needs **404** JSON, not a stack trace.

Remember: Two tables — `campus_products` and `campus_users`. MYSQL homework `users` is a different table. Do not drop it.

Interview: Walk the request: JSON → DTO → `@Valid` → mapper → entity → MySQL. Then say where JWT sits (**before** the controller on POST).

Print the happy path in one breath: GET menu public, register for a token, POST a snack with Bearer.

```
public class HappyPathBreath {
    public static void main(String[] args) {
        System.out.println("GET menu public");
        System.out.println("register -> token");
        System.out.println("POST snack with Bearer");
    }
}
```

Expected Output:

```
GET menu public
register -> token
POST snack with Bearer
```

Print that `ProductRequest` is in, `ProductResponse` is out, and `Product` stays the entity.

```
public class JsonNotTheRow {
    public static void main(String[] args) {
        System.out.println("ProductRequest in");
        System.out.println("ProductResponse out");
        System.out.println("Product stays the entity");
    }
}
```

Expected Output:

```
ProductRequest in
ProductResponse out
Product stays the entity
```

3. Feature freeze — IN vs OUT

Definition: **IN** items must work on demo day. **OUT** items are Stretch or a later module. Saying “not in freeze” in the README is **professional**, not lazy.

Purpose: Reviewers grade a **contract**. An unbounded scope never demos.

IN (must demo)	OUT (not this lab)
MySQL catalog + `campus_users`	Deploy, Actuator, PaaS (**Later —** next file)
DTOs + `@Valid` + lite **400** / **404**	Servlet vs Spring map (**Later —** module 08)
`POST /auth/register` **201**, duplicate **409**	OAuth2, refresh tokens, admin-role matrix
`POST /auth/login` **200** / **401**	React storefront / login UI
`GET /products` **200** without token	CORS required to pass (Stretch / **Later —**)
`GET /products/{id}` **200** / **404**	HTTP **DELETE**, extra catalog fields
`POST /products` **401** no token; **201** Bearer	Lombok required
`POST /products` Bearer + blank name → **400**	Kit 20, BW07
BCrypt in MySQL; JWT secret from **env**	Real secret or DB password in git
Postman kit 19 collection + README bundle	New Boot app besides kit 19

```
public class FreezeInOut {
    public static void main(String[] args) {
        System.out.println("IN: GET public POST JWT 400 404");
        System.out.println("OUT: deploy OAuth2 React DELETE");
    }
}
```

Expected Output:

```
IN: GET public POST JWT 400 404
OUT: deploy OAuth2 React DELETE
```

Note:
- Wrong: promising cart / orders / admin on this demo — capstone milestones are **Later —**.
- Wrong: skipping 400 because “auth was the last class.”
- Prefer: tick the IN table in the README so a peer can grade you.

Common Error: Adding `@CrossOrigin` and calling CORS “done.” CORS is Stretch. The freeze does not need a browser.

Remember: Stretch may exist in the repo. It must not block **Done when**.

Interview: How do you stop scope creep? Publish an IN/OUT list **before** coding the last night.

Print the freeze-in contract: catalog DTO Valid JWT, GET public POST protected, 400 with token and 401 without.

```
public class FreezeInPrint {
    public static void main(String[] args) {
        System.out.println("catalog DTO Valid JWT");
        System.out.println("GET public POST protected");
        System.out.println("400 with token 401 without");
    }
}
```

Expected Output:

```
catalog DTO Valid JWT
GET public POST protected
400 with token 401 without
```

Print freeze-out: no Actuator this lab, no OAuth2 refresh roles, no React required.

```
public class FreezeOutPrint {
    public static void main(String[] args) {
        System.out.println("no Actuator this lab");
        System.out.println("no OAuth2 refresh roles");
        System.out.println("no React required");
    }
}
```

Expected Output:

```
no Actuator this lab
no OAuth2 refresh roles
no React required
```

---

4. Must-pass statuses — 401 before 400

Definition: Must-pass is the smallest status set a peer can check in Postman without reading your source. Kit 19's collection is that checklist.

Purpose: "It works on my machine" is not a demo. **401 / 400 / 404 / 201 / 409** are the language of the freeze.

The order trap (this lab's one new emphasis): the filter chain runs **before** the controller. `@Valid` never sees a request that Security already rejected.

Call	Token	Body	Status	Why
`GET /products`	none	—	**200**	Public menu
`GET /products/99999`	none	—	**404**	Public read; missing row
`POST /products`	none	valid JSON	**401**	Not authenticated

Call	Token	Body	Status	Why
`POST /products`	none	`"name":""`	**401**	Still not authenticated — not 400
`POST /products`	Bearer	valid JSON	**201**	AuthN + AuthZ + persist
`POST /products`	Bearer	blank name	**400**	Known user; validation failed
`POST /auth/register`	none	new email	**201**	Public auth
`POST /auth/register`	none	same email	**409**	Unique email
`POST /auth/login`	none	wrong password	**401**	Failed AuthN

Lite 400 body (module 04 — **do not rewrite** the advice class):

```
{"error":"Validation failed","fields":{"name":"must not be blank"}}
```

Lite 401 body (module 05 entry point):

```
{"error":"Unauthorized"}
```

Note:
- Wrong: expecting 400 on anonymous POST with a blank name.
- Wrong: treating 403 as the core-path miss — this course uses **401** for anonymous / bad JWT. Role **403** is **Later —**.
- Prefer: run **Create product without token** before **Create product invalid**.

Common Error: Postman **Create invalid** still has no Authorization header after you protected POST — you will keep seeing 401 and think `@Valid` is broken.

Remember: 401 = door. 400 = form. 404 = unknown id. 201 = created. 409 = email taken.

Interview: Why can a blank-name POST return 401? Because AuthN failed first.

Print door-before-form: no Bearer is 401, Bearer plus blank name is 400, and `@Valid` never ran on 401.

```
public class DoorBeforeForm {
    public static void main(String[] args) {
        System.out.println("no Bearer -> 401");
        System.out.println("Bearer + blank name -> 400");
        System.out.println("@Valid never ran on 401");
    }
}
```

Expected Output:

```
no Bearer -> 401
Bearer + blank name -> 400
@Valid never ran on 401
```

Print public GET `/products/99999` with no token and 404 lite JSON.

```
public class Public404Print {
    public static void main(String[] args) {
        System.out.println("GET /products/99999");
        System.out.println("no token needed");
        System.out.println("404 lite JSON");
    }
}
```

Expected Output:

```
GET /products/99999
no token needed
404 lite JSON
```

---

5. Peer break-auth — diagnose, do not invent attacks

Definition: Break-auth means the API **claims** to be protected but a guest can still write, or a password is stored in the clear, or a secret is in git. Kit 19 **`broken/`** is the classroom patient. A peer's clone may show the same smells.

Purpose: Interviews ask "how would you test auth?" You answer with **four checks**, not a new exploit chapter. Fixes are already in module 05 + kit `solution/` (trainer-only in the student pack).

Smell in `broken/`	What you see	Fix already taught
Raw password / `NoOpPasswordEncoder`	Workbench shows `IdliDosa8`; hash does not start `$2`	`BCryptPasswordEncoder` + `encode` / `matches`
Filter never reads `Authorization`	Header present but POST still anonymous **or** fake token accepted	`JwtAuthFilter` + `SecurityContext` (module 05)
`anyRequest().permitAll()`	POST **201** with **no** Bearer	`GET` permitAll; `anyRequest().authenticated()`
Secret / DB password literals in yml	`git grep` finds `root123` or a committed JWT string	`${JWT_SECRET}` / `${MYSQL_PASSWORD}`; commit `.env.example` only
Fake JWT (`fake.email.token`)	Token is not three signed parts	`jjwt` builder + env secret (module 05)

Do not add XSS, CSRF-cookie labs, SQL-injection chapters, or brute-force tooling. JDBC already used `PreparedStatement`. CSRF is off **on purpose** for this Bearer API (module 05). Deploy hardening is **Later —**.

Peer script (speak it, then click Postman):

1. POST a snack **without** Bearer — must be **401**.
2. Register — Workbench `password_hash` starts with **`$2`**.
3. `git grep` / open yml — no real password, no literal JWT secret.
4. If POST is 201 with no header → look for **`permitAll`**.

```
public class FourSmellsPrint {
    public static void main(String[] args) {
        System.out.println("plain password");
```

```
        System.out.println("missing Bearer filter");
        System.out.println("permitAll everywhere");
        System.out.println("secret in git");
    }
}
```

Expected Output:

```
plain password
missing Bearer filter
permitAll everywhere
secret in git
```

Note:
- Wrong: rewriting `broken/` into a new security framework.
- Wrong: committing the peer's real `.env` "so I can help."
- Prefer: name the smell in one sentence, point at the module 05 fix, re-run the collection.

Common Error: "I added `permitAll` so Postman would work." That **is** the bug the peer should catch.

Remember: A 201 without a token is not a green test. It is a failed freeze.

Interview: How do you know passwords are hashed? Select the column. `$2a$` / `$2b$` / `$2` prefix — never `IdliDosa8`.

Print the permitAll smell: POST `/products` with no header — 201 is a bug, 401 means the door works.

```
public class PermitAllSmell {
    public static void main(String[] args) {
        System.out.println("POST /products no header");
        System.out.println("201 means permitAll bug");
        System.out.println("401 means the door works");
    }
}
```

Expected Output:

```
POST /products no header
201 means permitAll bug
401 means the door works
```

Print git rules: `.env.example` yes, JWT_SECRET literal no, YAML uses `${JWT_SECRET}`.

```
public class SecretGrepPrint {
    public static void main(String[] args) {
        System.out.println("git: .env.example YES");
        System.out.println("git: JWT_SECRET literal NO");
        System.out.println("yml uses ${JWT_SECRET}");
    }
}
```

Expected Output:

```
git: .env.example YES
git: JWT_SECRET literal NO
yml uses ${JWT_SECRET}
```

---

6. Demo day — deliverables bundle

Definition: The **bundle** is what a peer clones. Same checklist as Standard Project Deliverables — except **no live URL**. Deploy is **Later —** the next lab.

Purpose: A 2–5 min video that only shows `mvn spring-boot:run` is not a freeze demo. Show **Guest GET**, **401 POST**, **register**, **201 POST**, **400**, **404**. README must list env keys.

Artefact	This lab
Public GitHub repo	No secrets; `target/` ignored
`README.md`	Freeze IN/OUT, setup, curl / Postman, screenshots
`pom.xml`	Boot 3, JDK 17, `jakarta.*` starters already in kit 19
`.env.example`	`MYSQL_*`, `JWT_SECRET` dummy (≥ 32 chars)
Seed / sample	Document `training_db`; `ddl-auto: update` creates tables
Screenshots	Postman 201 + 401 + Workbench `$2` hash (crop secrets)
Demo video 2–5 min	Walk the must-pass table
`.gitignore`	`target/`, `.env`, `*.class`, `.idea/`
Postman	Kit 19 `CampusCart_Auth_v1.postman_collection.json`

Sample credentials in README: `priya@campuscart.in` / lab password **from** `.env.example` **notes** — not a production password. Never paste a live JWT into the README.

Note:
- Wrong: screenshot of a token that is still valid — crop or expire it.
- Wrong: “Live: coming soon” as if deploy were required.
- Prefer: “Deploy is the next lab. This repo is localhost + Postman.”

Common Error: Committing `application.yml` with `password: root123` “because the video needed it.” Rotate if it hits git.

Remember: Clone instructions that need a trainer’s brain are not independent.

Interview: What do you show in 3 minutes? Guest read → failed write → login → successful write → validation → missing id.

Print what README must list: export `MYSQL_PASSWORD` and `JWT_SECRET`, `mvn spring-boot:run`, import the Postman collection.

```
public class ReadmeMustPrint {
    public static void main(String[] args) {
        System.out.println("export MYSQL_PASSWORD JWT_SECRET");
        System.out.println("mvn spring-boot:run");
        System.out.println("import Postman collection");
    }
}
```

Expected Output:

```
export MYSQL_PASSWORD JWT_SECRET
mvn spring-boot:run
import Postman collection
```

Print demo order: GET 200, POST 401, register 201, then POST 201, POST 400, GET 404.

```
public class DemoOrderPrint {
    public static void main(String[] args) {
        System.out.println("GET 200");
        System.out.println("POST 401");
        System.out.println("register 201");
        System.out.println("POST 201 then 400 then GET 404");
    }
}
```

Expected Output:

```
GET 200
POST 401
register 201
POST 201 then 400 then GET 404
```

Step-by-Step Annotated Java Code

Work from 15_Lab_Assets/19_spring_auth_tasks/. Open the folder that contains **`pom.xml`**. Do not create kit 20. Do not start a second Boot app in this repo.

Step 1: Env + run (same as module 05)

Copy `.env.example` values into the **shell** (Boot does not load `.env` by itself). `JWT_SECRET` stays ≥ 32 characters.

```
public class EnvKeysPrint {
    public static void main(String[] args) {
        System.out.println("MYSQL_PASSWORD");
        System.out.println("JWT_SECRET");
        System.out.println("not in yml literals");
    }
}
```

Expected Output:

```
MYSQL_PASSWORD
JWT_SECRET
not in yml literals
```

From the kit root:

```
export MYSQL_PASSWORD='your_lab_password'
export MYSQL_USER=root
export MYSQL_DATABASE=training_db
export JWT_SECRET='change_me_lab_jwt_secret_32chars!'

cd starter          # or broken until it matches Done when
mvn spring-boot:run
```

Print the three beans that must start: Tomcat, DataSource, `SecurityFilterChain`.

```
public class WhatMustStart {
    public static void main(String[] args) {
        System.out.println("Tomcat");
        System.out.println("DataSource");
        System.out.println("SecurityFilterChain");
    }
}
```

Expected Output:

```
Tomcat
DataSource
SecurityFilterChain
```

Print Boot 3 imports `jakarta.validation.Valid` and `jakarta.persistence.Entity` — not `javax`.

```
public class JakartaNotJavax {
    public static void main(String[] args) {
        System.out.println("jakarta.validation.Valid");
        System.out.println("jakarta.persistence.Entity");
        System.out.println("Boot 3");
    }
}
```

Expected Output:

```
jakarta.validation.Valid
jakarta.persistence.Entity
Boot 3
```

Step 2: Prove the freeze with curl / Postman

Import `CampusCart_Auth_v1.postman_collection.json`. Variable `baseUrl` = `http://localhost:8080`. Run **Register** (or **Login**) **before** any Bearer POST — including the **400** request.

Second terminal (replace `PASTE_TOKEN`):

```
curl -s http://localhost:8080/products

curl -s -o /dev/null -w "%{http_code}\n" -X POST http://localhost:8080/products \
  -H "Content-Type: application/json" \
  -d '{"name":"Pani puri","priceInr":30}'
# expect 401

curl -s -X POST http://localhost:8080/auth/register \
  -H "Content-Type: application/json" \
  -d '{"email":"priya@campuscart.in","password":"IdliDosa8"}'

curl -s -X POST http://localhost:8080/products \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer PASTE_TOKEN" \
  -d '{"name":"Pani puri","priceInr":30}'
# expect 201

curl -s -o /dev/null -w "%{http_code}\n" http://localhost:8080/products/99999
# expect 404

curl -s -o /dev/null -w "%{http_code}\n" -X POST http://localhost:8080/products \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer PASTE_TOKEN" \
  -d '{"name":"","priceInr":30}'
# expect 400 — not 401
```

Print collection order: Register saves the token, 401 POST has no Authorization, 400 POST still has Bearer.

```
public class CollectionOrderPrint {
    public static void main(String[] args) {
        System.out.println("Register saves token");
        System.out.println("401 POST has no Authorization");
        System.out.println("400 POST still has Bearer");
    }
}
```

Expected Output:

```
Register saves token
401 POST has no Authorization
400 POST still has Bearer
```

Print that registering the same email twice returns 409, not 201.

```
public class Duplicate409Print {
    public static void main(String[] args) {
        System.out.println("same email twice");
        System.out.println("409 not 201");
    }
}
```

Expected Output:

```
same email twice
409 not 201
```

Step 3: Peer break-auth on `broken/` (or a partner clone)

Switch to `broken/` **or** swap laptops. Do **not** open `solution/` until you have named the smells (or the trainer says to).

Checklist to speak out loud:

1. POST without token → **201**? → `permitAll`.
2. Hash column = `IdliDosa8`? → raw password / `NoOp`.
3. Header set but still 401 on a "valid" fake token? → filter or `JwtService`.
4. yml contains `root123` or `super-secret-please-commit-me-ok`? → env.

Print that a BCrypt hash starts with `$2`, and storing `IdliDosa8` in MySQL is a fail.

```
public class HashPrefixPrint {
    public static void main(String[] args) {
        System.out.println("BCrypt starts with $2");
        System.out.println("IdliDosa8 in MySQL is a fail");
    }
}
```

Expected Output:

```
BCrypt starts with $2
IdliDosa8 in MySQL is a fail
```

Print that `fake.email.token` is not HS256, and jjwt signs with `JWT_SECRET`.

```
public class FakeTokenSmell {
    public static void main(String[] args) {
        System.out.println("fake.email.token is not HS256");
        System.out.println("jjwt signs with JWT_SECRET");
    }
}
```

Expected Output:

```
fake.email.token is not HS256
jjwt signs with JWT_SECRET
```

Step 4: Bundle for demo day

Copy the README template in Standard Project Deliverables. Paste your freeze IN/OUT table. Add screenshots of Postman 201 and 401. Link the video. Push **without** `.env`.

Print the freeze ignore list: `target/`, `.env`, `*.class`.

```
public class GitIgnoreFreeze {
    public static void main(String[] args) {
        System.out.println("target/");
        System.out.println(".env");
        System.out.println("*.class");
    }
}
```

Expected Output:

```
target/
.env
*.class
```

Print what not to promise: no live PaaS URL, no React login, next lab is deploy.

```
public class WhatNotToPromise {
    public static void main(String[] args) {
        System.out.println("no live PaaS URL required");
        System.out.println("no React login");
        System.out.println("next lab is deploy");
    }
}
```

Expected Output:

```
no live PaaS URL required
no React login
next lab is deploy
```

Practice Exercises

(Each Practice item includes a filled Model answer: and Expected Output: where the demo prints.)

Basic

1. Print what consolidation glues.

Model answer:

```
public class WhatGluePrint {
    public static void main(String[] args) {
        System.out.println("JPA catalog");
        System.out.println("DTO Valid 400 404");
        System.out.println("JWT door on POST");
    }
}
```

Expected Output:

```
JPA catalog
DTO Valid 400 404
JWT door on POST
```

2. Print the public vs protected rule.

Model answer:

```
public class PublicVsProtected {
    public static void main(String[] args) {
        System.out.println("GET /products public");
        System.out.println("POST /products JWT");
    }
}
```

Expected Output:

```
GET /products public
POST /products JWT
```

3. Print where the JWT secret lives.

Model answer:

```
public class SecretLivesPrint {
    public static void main(String[] args) {
        System.out.println("environment JWT_SECRET");
        System.out.println("not a yml literal in git");
    }
}
```

Expected Output:

```
environment JWT_SECRET
not a yml literal in git
```

Intermediate

1. Why blank-name POST without a token is 401.

Model answer:

```
public class BlankName401 {
    public static void main(String[] args) {
        System.out.println("SecurityFilterChain first");
        System.out.println("anonymous -> 401");
        System.out.println("@Valid did not run");
    }
}
```

Expected Output:

```
SecurityFilterChain first
anonymous -> 401
@Valid did not run
```

2. Why GET by id 404 stays public.

Model answer:

```
public class GetByIdPublic {
    public static void main(String[] args) {
        System.out.println("menu detail is still a read");
        System.out.println("missing id is 404 not 401");
    }
}
```

Expected Output:

```
menu detail is still a read
missing id is 404 not 401
```

3. Name the four break-auth smells.

Model answer:

```
public class NameFourSmells {
    public static void main(String[] args) {
        System.out.println("plain password or NoOp");
        System.out.println("filter skips Bearer");
        System.out.println("permitAll on POST");
        System.out.println("secret in git");
    }
}
```

Expected Output:

```
plain password or NoOp
filter skips Bearer
permitAll on POST
secret in git
```

4. What is out of freeze.

Model answer:

```
public class OutOfFreezePrint {
    public static void main(String[] args) {
        System.out.println("deploy Actuator");
        System.out.println("OAuth2 refresh roles");
        System.out.println("React DELETE CORS Lombok required");
    }
}
```

Expected Output:

```
deploy Actuator
OAuth2 refresh roles
React DELETE CORS Lombok required
```

Interview-Oriented

1. What is a feature freeze?

Model answer: A locked IN list for demo day. Extra fields, DELETE, CORS, and Lombok may exist as Stretch. They are not required to pass.

2. Does this lab replace module 05?

Model answer: No. It proves JWT + BCrypt + `SecurityFilterChain` still guard POST while the catalog DTO path from 03–04 still returns 400/404.

3. 401 vs 400 vs 404 on this API?

Model answer: 401 = no/invalid JWT (or bad login). 400 = `@Valid` failed on a request that already passed the door. 404 = unknown product id on a public GET.

4. How do you test auth without attacking anyone?

Model answer: POST without Bearer (expect 401), check BCrypt prefix in MySQL, `git grep` for secrets, look for `permitAll` on writes. Kit `broken/` holds those smells.

5. Is a deploy URL required?

Model answer: No. The bundle is README + Postman + video on localhost. Actuator / PaaS is the next protected lab.

6. Why `jakarta.*`?

Model answer: Spring Boot 3. `javax.*` Bean Validation / Servlet packages are the old stack.

Mini-Project (Assignment): CampusCart CRUD + Auth Freeze

Lab kit: 15_Lab_Assets/19_spring_auth_tasks/

Modules 03–05 already taught persist, DTOs, and JWT lite. This assignment is the **portfolio freeze**: one CampusCart API a peer can clone, run, and break-auth. Push a public repo. No secrets in git. No kit 20.

Do **not** add OAuth2, refresh tokens, role matrices, deploy, or Lombok to pass. Extra fields, DELETE, and CORS are Stretch.

What to build

Piece	Detail
App	Kit 19 Maven tree (`starter/` TODOs or `broken/` fixes) — JDK 17, Boot 3, `jakarta.*`
Persistence	Keep **Product** DTO GET list / GET by id / POST on MySQL (`campus_products`)

Piece	Detail
Validation	`@Valid` on POST; lite **400** `{error, fields}`; **404** on missing id
Auth	**POST** `/auth/register` **201**; duplicate **409**; **POST** `/auth/login` **200** / **401**
Hash	BCrypt in `campus_users` — not the raw password
Door	GET `/products` and GET `/products/{id}` **permitAll**; POST **authenticated**
Token	`Authorization: Bearer`; secret `${JWT_SECRET}` (≥ 32 chars)
Postman	Kit 19 collection — including 401, 201, **400 with Bearer**, **404**, duplicate **409**
Bundle	README freeze IN/OUT, `.env.example`, screenshots, 2–5 min demo (Standard Project Deliverables — **no** live URL)
Secrets	`.env.example` only; `.gitignore` has `target/` and `.env`

Work from `starter/` or `broken/`. Compare `solution/` only after you finish (or when told in class). `solution/` is trainer-only in the student pack.

Stretch

- Extra catalog fields (description, veg flag) — **not** Done when.
- Authenticated **DELETE** `/products/{id}` — optional.
- `@CrossOrigin(origins = "http://localhost:5173")` — **not** Done when. React `fetch` is **Later —**.
- Lombok / records — optional, not in Done when.
- **Later —** deploy / Actuator (next module). **Later —** OAuth2, refresh, admin roles. **Later —** Servlet vs Spring map.

Done when

- [ ] `mvn spring-boot:run` starts (Tomcat + DataSource + Security)
- [ ] Register **201** with `token` + `email` (no password field); hash in MySQL starts with `$2`
- [ ] Login **200** / bad password **401**; duplicate email **409**
- [ ] `GET /products` **200** without token
- [ ] `GET /products/{id}` **200** existing / **404** missing (lite JSON)
- [ ] `POST /products` **401** without token; **201** with Bearer
- [ ] `POST /products` with Bearer + blank name → **400** (`error` + `fields`) — not 401
- [ ] Feature freeze OUT respected (no deploy / OAuth2 / refresh / roles / React / required DELETE / CORS / Lombok)
- [ ] Peer break-auth: can name the four smells (plain password, missing Bearer, `permitAll`, secret in git) and the fix
- [ ] Kit 19 Postman collection imports; statuses match the rows above
- [ ] `JWT_SECRET` and DB password are not literals in git
- [ ] README + screenshots + 2–5 min demo; `target/` untracked; no live PaaS URL required

Interview Q&A

Shared with students after concepts and the mini-project. Use for review and interview prep — questions and model answers are included.

Q1. What did you freeze?

A: One CampusCart API: MySQL catalog with DTOs and `@Valid`, lite 400/404, JWT register/login, public GET, protected POST. Not deploy, not OAuth2, not React.

Q2. Why is this a lab, not a new Security chapter?

A: BCrypt, `SecurityFilterChain`, and JWT were module 05. JPA was 03. DTOs and Postman were 04. This lab proves they still work in one process.

Q3. Why 401 on a blank-name POST with no token?

A: Filters run first. Anonymous callers never reach `@Valid`. Add Bearer, then the same body becomes 400.

Q4. 401 vs 403 vs 400 vs 404?

A: 401 = not authenticated. 403 = authenticated but forbidden (roles are later). 400 = validation. 404 = no product for that id.

Q5. How do you break-auth a peer without hacking?

A: POST with no header (expect 401), inspect the hash column, search git for secrets, read `SecurityFilterChain` for `permitAll` on writes.

Q6. Why keep GET public after adding JWT?

A: A campus menu is a read. Writes need a known user. Hiding GET is a product choice, not the core-path freeze.

Q7. Where do secrets live?

A: Environment variables. Commit `.env.example` with dummy values. `application.yml` uses `${MYSQL_PASSWORD}` and `${JWT_SECRET}`.

Q8. Does JWT replace JPA?

A: No. The same `Product` row is saved. Security only decides **who** may call POST.

Q9. Why `jakarta.*`?

A: Spring Boot 3. Validation and persistence annotations are `jakarta.*`, not `javax.*`.

Q10. Is CORS part of Done when?

A: No. Stretch only. Curl and Postman are not browsers. React weeks need CORS later.

Q11. What belongs in the demo video?

A: GET 200, POST 401, register, POST 201, POST 400 with token, GET 404, Workbench `$2` hash. Mention IN/OUT in the README.

Q12. Why not Lombok to pass?

A: Juniors must show mapping and getters. Lombok is optional Stretch.

Q13. How does this connect to CampusCart capstone?

A: This freeze is the catalog + auth shell. Cart, orders, admin, and deploy are later milestones.

Q14. What if POST returns 201 with no token?

A: The chain likely `permitAll()`s everything. That is a failed freeze, not a successful create test.

Q15. Why no kit 20?

A: Kit 19 already has catalog + auth. A second tree would split the demo. Stretch stays in the same repo.

Next

Push **CampusCart CRUD + Auth Freeze** (kit 19 — collection committed, no secrets). Continue with **Lab: Deploy + Capstone Debug** — Actuator / health + env. OAuth2, refresh tokens, admin roles, and React login stay **Later —**.